Midterm project

18/10/2026

30 Points Possible

Attempt 1

In Progress

NEXT UP: Submit assignment

Add comment

Unlimited Attempts Allowed

12/10/2026 to 20/10/2026

Details

Midterm project: CryptoZombies DApp

For midterm project, you are to wrap up what you have learned from CryptoZombies lessons and build a DApp.

This DApp should at least contain backend smart contracts and a frontend website.

The minimum requirement is to run the provided starter code package at a local machine.

You should run this DApp at least on a local blockchain network, Ganache.

The following video will show you how to meet this minimum requirement.

Download starter code package here! (https://csufullerton.instructure.com/courses/3586622/files/284230861?wrap=1)

The starter code is on the following versions:
Truffle v5.4.25
Ganache v2.5.4
Solidity - 0.8.11 (solc-js)
Node v14.16.0
Web3.js v1.2.7

Further improve the DApp will land you with a better grade.

The grading is subjective and is based on the overall performance and how it's compared to your peers'.

To receive points on a certain improvement, you'll need to demo the improvement.

You don't need to demo how to install, deploy and setup due to limited time.

Deploy contracts before the demo and prepare a ready-to-showcase version.

Some suggestions for further improving the starter code:

1: Deploy to other testnets, for example, Goerli, etc.

2: Build a nicer web site.

3: Add more js code to demo other functionalities, for example, feed on kitties, transfer, etc.

4: Add kitty smart contract to the backend.

5: Add images to demo zombies in a better way.

6: Build a Zombie factory/army and demo multiple Zombies.

7: A battle game of Zombie vs Kitty.

8: NFT marketplace.

9: Change the starter code to NOT hard-coded.

10: And many more!

You and your team will need to demo your DApp next week, in person in class.

In-person attendance and sign-up are required unless excused due to medical or other emergency situations.

Plz. limit your demo to maximum 10 minutes .

You will need to email me by the end of this week if you cannot attend due to a non-emergency reason.

You will need to email me within 10 days after the next Monday if you cannot attend due to an emergency situation.

Plz. bring your own device (laptop, hdmi adaptor, etc.) for the demo.

If your team members are from **difference sections**, you can choose from one of the following options:
1) Discuss with your teammates to see if all of you can go to the same section for the demo. This option is preferred.
2) If all the members cannot make to the same section, pick a section that fit most of the members' schedule.
For example, 4 members in a team, 3 members can go to Section 1, then these 3 members will do the demo at Section 1.
The fourth member does NOT need to do another demo, but need to show up at any section to sign up on the attendance sheet.

Deliverables:

Write a readme file (text file, any format) to include all team member name, CWID, email.

In this readme file, explain any improvement that you have made.

If there is no improvement, just write "no further improvement".

If the DApp is significantly different from the starter code, plz. also include instructions for how to set it up and run.

Turn in your DApp, video demo, slides if any in one package in a .rar file.

If your DApp is on Github, you can include a repo link in the readme file instead.

One team only submit ONE copy.

You can submit multiple times before the deadline, but only the latest submission will be graded.

Pick a team member who will be responsible for submitting the project.
Other members do not need to submit. Canvas may mark your project as "Missing", but you do not need to worry about it.
I'll contact you if I cannot find your name in any readme files or team sign-up sheet.

You can find the rubric below but this rubric is only for general guidance.

The grading is based on a subjective manner since it is an open-ended project.

References:

Zombie factory GitHub repo

https://github.com/loomnetwork/cryptozombies-lesson-code (https://github.com/loomnetwork/cryptozombies-lesson-code)

Set up a local http server

https://www.npmjs.com/package/http-server (https://www.npmjs.com/package/http-server)

Troubleshooting:

1) Running starter code package:

Make sure ganache settings (url, port, account address), truffle-config.js, index.html contract address, metamask account address, metamask network settings MATCH CONSISTENTLY.

Check consistency whenever one of them changes!

You should use **ZombieOwnership** contract address, NOT ZombieFactory or any other smart contract addresses.

2) Kitty contract

In CryptoZombie lesson 2, Chapters 11-14, it uses **the REAL Cryptokitty contract address**.
So there is only a Kitty interface created.
However, the real kitty contract does not allow "eating" their kitties due to security consideration.
If you are to implement it, you will need to create your own kitty contract locally.
You will need to implement necessary functions, not only an interface.
Refer to the Cryptokitty smart contract code in the following:
https://etherscan.io/address/0x06012c8cf97bead5deae237070f9587f8e7a266d#code
(https://etherscan.io/address/0x06012c8cf97bead5deae237070f9587f8e7a266d#code)
https://etherscan.io/address/0xb1690c08e213a35ed9bab7b318de14420fb57d8c#code
(https://etherscan.io/address/0xb1690c08e213a35ed9bab7b318de14420fb57d8c#code)

View Rubric

Rubric for midterm project

Criteria	Ratings					Points
Meet minimal requirement view longer description	**Full Marks** demo start code package without any issue. **24 pts**	**demo start code package with some minor issues** **20 pts**	**demo start code package with some major issues** **16 pts**	**meet partial requirement** submitted code package but did not demo in class and absence is not excused; demo in class but did not submit code package **8 pts**	**did not submit and fail to demo/run start code package in class** **0 pts**	/24 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts
Improvement view longer description	**Full Marks** add and demo one improvement **1 pts**			**No Marks** did not add and demo one improvement **0 pts**		/1 pts